A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology

Daniel J. Eck[1], Shen Yan[1*]

1. Department of Statistics, University of Illinois Urbana-Champaign

Abstract

We study a paradox in which an individual with greater latent aptitude exhibits lower observed achievement than an individual with lesser aptitude when comparisons are made across contexts. We formalize this phenomenon as the aptitude-achievement paradox and introduce its probability as an explicit inferential target. To study this quantity, we employ Full House Modeling, a framework that derives latent aptitude from observed achievement by embedding observed samples within larger source populations. We derive closed-form paradox probabilities when achievement distributions are identical across contexts and obtain explicit expressions and bounds when they differ, including tractable results under normality with reliable plug-in estimation. Our results show that paradox risk depends primarily on population and sampling ratios rather than absolute sizes. An application to Major League Baseball contextualizes historically extreme batting performances, showing how modern achievements may appear less exceptional after accounting for population depth and measurement environments. By elevating aptitude-achievement reversals from anecdotal concern to an estimable

*Daniel J. Eck (dje13@illinois.edu); email addresses: Shen Yan (shenyan3@illinois.edu)

quantity, this work provides a principled foundation for fairer and more interpretable cross-context comparisons.

1 Introduction

Comparisons of performances across differing contexts, schools, regions, eras, institutions, and tasks are central to education and psychometrics [Holland and Wainer, 2012, Bond and Fox, 2013], labor economics [Nickell and Layard, 1999], medicine, and sports analytics [Berry et al., 1999, Schell, 2013, 2016, Winston et al., 2022, Yan et al., 2025]. These problems have led to the development of a variety of widely used modeling frameworks that estimate relative ability, including empirical-Bayes shrinkage [Efron and Morris, 1975], Bayesian hierarchical models [Raudenbush and Bryk, 1986, Berry et al., 1999], Bradley-Terry-Luce and paired comparisons models [Bradley and Terry, 1952, Bradley, 1976, 1984, David, 1988], Elo-type dynamic rating systems [Elo, 1978, Graepel et al., 2007], Gaussian-process and state-space formulations [Chu and Ghahramani, 2005, Houlsby et al., 2012], item-response and latent class models [McDonald, 1999, Embretson and Reise, 2000, Hagenaars and McCutcheon, 2002, Holland and Wainer, 2012, Bond and Fox, 2013], factor models [McDonald, 1999, Bartholomew et al., 2011], and domain-specific metrics [Baumer and Matthews, 2014, Baumer et al., 2015, Yu et al., 2016].

Despite their differences, these approaches share a common structural assumption: ability is inferred relative to a fixed or implicitly comparable set of observed competitors. While such models adjust outcomes within a given context, they do not explicitly account for changes in the size or structure of the eligible population, nor for context-specific distortions in how latent ability is transformed into observed outcomes. As a consequence, they may not be able to distinguish observed dominance from underlying aptitude when competition depth, selection mechanisms, or measurement conditions vary across contexts.

These limitations give rise to a previously unquantified inferential target: the probabil-

ity that cross-context comparisons reverse latent orderings. Specifically, an individual drawn from a deeper or more competitive population may possess higher latent aptitude yet exhibit lower observed achievement than a counterpart from a shallower or more weakly filtered context. We term this phenomenon the aptitude-achievement paradox. Formally, let individual j in context i have an observed $\mathbb{R}$-valued statistic $Y_{i,j}$ and an underlying latent $\mathbb{R}$-valued aptitude $X_{i,j}$, and let individual n in context m have an observed $\mathbb{R}$-valued statistic $Y_{m,n}$, and an underlying latent $\mathbb{R}$-valued aptitude $X_{m,n}$. The probability of aptitude-achievement paradox is

$$\mathrm{P}\left(X_{i,j} > X_{m,n}, Y_{i,j} < Y_{m,n}\right) \tag{1}$$

This quantity captures the structural risk that cross-context comparisons based on observed outcomes reverse the true latent ordering.

The aptitude-achievement paradox in Eq 1 cannot arise if a single, strictly increasing, context-invariant measurement map relates aptitude to achievement and if selection into tasks is ignorable. In practice, however, both measurement heterogeneity (non-invariance, differential item functioning) and non-ignorable selection (who attempts which tasks, faces which opponents, receives which treatment) are pervasive. These features can reverse cross-context rankings even when the latent ranking is unambiguous, echoing classic reversal phenomena such as Simpson's [Simpson, 1951] and Lord's paradoxes [Lord, 1967] while implicating modern concerns about fairness. What practitioners lack is a principled way to quantify how often such reversals should occur under an explicit model of aptitude, measurement, and selection.

A large class of methods quantifies how often observed orderings disagree with a target ordering, closely related to the phenomenon studied here. Classic examples include rank-discordance measures such as Kendall's τ [Kendall, 1938] and Somers' D [Somers, 1962], which summarize the frequency of concordant and discordant pairs based on observed rankings. Because these indices operate entirely on observed outcomes, they neither place individuals on a common latent scale nor disentangle the effects of measurement non-invariance

or selection, motivating models that make the measurement and selection processes explicit.

Traditional latent trait models [Bartholomew et al., 2011] focus on estimating underlying abilities from observed categorical or discrete outcomes. For example, the Rasch model [Bond and Fox, 2013] ensures that comparisons of individuals' abilities are invariant to the specific items administered, under the assumption of no differential item functioning (DIF) [Holland and Wainer, 2012]. However, such models do not explicitly leverage information about the size or structure of the source population from which individuals are drawn. Likewise, Bayesian hierarchical and state-space models for latent ability—including Elo-type rating systems [Berry et al., 1999, Elo, 1978, Graepel et al., 2007]—typically condition on the observed competitive or testing pool and capture population-level variation through random effects or dynamic evolution, without modeling variation in eligible population depth or selection into observation across contexts. In parallel, the causal bootstrap [Imbens and Menzel, 2018] emphasizes that valid causal inference requires explicitly situating the observed sample within a larger source population, rather than treating the realized sample as the population of interest.

To address these limitations, we develop a framework for cross-context comparison that explicitly accounts for population depth and selection mechanisms. This framework, which we refer to as Full House Modeling (FHM), was initially proposed in an applied setting comparing baseball players across eras [Yan et al., 2025], and is developed here as a general approach for cross-contextual evaluation. This framework separates latent aptitude from observed achievement by modeling how three features jointly shape observed rankings: (i) the size of the eligible population in each context, (ii) the mechanism by which individuals are selected or observed, and (iii) the context-specific mapping from latent aptitude to observable performance. Under FHM, individuals are drawn from a common latent ability distribution, while contexts differ in population depth and in how aptitude is filtered and measured. This formulation makes explicit the structural mechanisms through which observed performance can be inflated or suppressed independently of true underlying aptitude.

We model measurement as a rank-preserving transformation of latent aptitude within each context.

Full House Modeling is a general framework. Its assumptions are most naturally satisfied in settings characterized by strong selection into the observed system. In elite competition environments, where observed individuals represent the upper tail of a larger latent population, selection can often be credibly approximated by top-n inclusion rules, yielding tractable theoretical results and clean empirical interpretability. In less selective settings, the selection mechanism requires more careful specification, and we regard the development of FHM under richer selection models as an important direction for future work. Comparisons of athletes across eras provide a canonical illustration of the favorable regime: entry into professional leagues is highly competitive and closely approximates selection of the most talented individuals from a large underlying population, making the top-n assumption both transparent and defensible. Our application to Major League Baseball is motivated by this alignment between the framework's assumptions and the structure of the domain.

In this article, we formalize Full House Modeling and study the probability of aptitude–achievement paradoxes (1) arising in cross-context comparisons. We derive explicit paradox probabilities in two settings: first, when observed achievement is independent and identically distributed, and second, when observed achievement follows independent but context-specific normal distributions. We also establish bounds on paradox probabilities in more general settings and show that, under normality, these probabilities can be consistently estimated using plug-in methods. A sensitivity analysis reveals that paradox risk depends on ratios of eligible population sizes rather than their absolute magnitudes. Proofs of mathematical results are provided in online Supplementary Materials.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We distinguish between a *population*, consisting of all individuals N_i in context i, and a *system*, which denotes the subset of individuals whose outcomes are observed or recorded. The system is determined by a selection mechanism that maps the full population into an observed sample of size n_i. We suppose that every individual $k = 1, \ldots, N_i$ has an underlying latent $\mathbb{R}$-valued aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{iid}{\sim} F_X$. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j. We formalize the selection mechanism through an indicator function $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right)$, where a value of 1 indicates inclusion in the observed system and a value of 0 indicates exclusion. This function may depend on the individual's aptitude relative to others in the population.

We now suppose that an individual j included in system i has an observed $\mathbb{R}$-valued statistic $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$, with $n_i \leq N_i$. Under Full House Modeling, the observed values $Y_{i,j}$ are assumed to preserve the ranking induced by the latent aptitudes among the selected individuals.

Let $X_{i,(j)}$ and $Y_{i,(j)}$ denote the jth order statistic for system i. Here, larger values correspond to higher aptitude or achievement, so that $X_{i,(j)} > X_{i,(k)}$ whenever $j > k$. We form (Y, X) pairs of random variables for each individual included in the system. These pairs are of the form $(Y_{i,(j)}, X_{i,(r_{i,j})})$, where $r_{i,j}$ indexes the rank of that individual's latent aptitude within the full population of size N_i.

As an example, consider the selection rule $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$, which selects the top n_i individuals in population i. In this case, the pairs take the form $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ for $j = 1, \ldots, n_i$.

The key feature of Full House Modeling is the explicit connection between the elements forming the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs. This construction can be understood as a rank-preserving transformation between latent aptitude and observed achievement. Based on the probability

integral transformation, Full House Modeling specifies this connection as

$$\begin{aligned} Y_{i,(j)} &= F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}} \left(X_{i,(r_{i,j})} \right), \\ X_{i,(r_{i,j})} &= F^{-1}_{X_{i,(r_{i,j})}} \circ F_{Y_{i,(j)}} \left(Y_{i,(j)} \right), \end{aligned} \tag{2}$$

where $1_i(X_{i,(r_{i,j})}, \mathbf{X}_{i,-(r_{i,j})}) = 1$, and both F_{Y_i} and F_X are continuous and invertible. We summarize Full House Modeling in the following definition.

Definition 2.1 (Full House Model)**.** *Consider populations with sizes N_i and system sizes n_i, $N_i \geq n_i$. Let a system inclusion function $1_i(\cdot, \cdot)$ be specified. Let $X_{i,j} \overset{iid}{\sim} F_X$, $j = 1, \ldots, N_i$. Let $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$. Form pairs $(Y_{i,(j)}, X_{i,(r_{i,j})})$, $j = 1, \ldots, n_i$, for the individuals included in the system, that is, those for which $1_i(\cdot, \cdot) = 1$. Then the Full House Model generates $Y_{i,(j)}$ values following (2) where F_{Y_i} and F_X are continuous and invertible.*

The Full House Model defines a joint distribution over (X, Y) together with a selection mechanism $1_i(\cdot, \cdot)$ as follows. Latent variables X are drawn i.i.d. from F_X, and the selection mechanism $1_i(X_{i,j}, \mathbf{X}_{i,-j})$ determines inclusion in the observed system. Conditional on inclusion, observed outcomes Y are generated via a rank-preserving monotone coupling with X. In particular, the dependence between X and Y is induced through a quantile-matching transformation of their order statistics, ensuring that higher latent aptitude corresponds to higher observed achievement within each system. The stochasticity in the Full House Model arises entirely from the latent population X, while both selection and observed outcomes are deterministic functionals of X.

The Full House Model induces structural missingness by construction: for any individual j such that $1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$, the corresponding observed statistic $Y_{i,j}$ is undefined. In subsequent sections, we consider probabilities involving both latent and observed quantities. To formalize this setting, when $1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$, we treat $Y_{i,j}$ as not defined within the observable probability space. For example,

$$\mathbb{P}\big(X_{i,j} \in A,\ Y_{i,j} \in B,\ 1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 1\big)$$

is well-defined for Borel sets $A, B \subset \mathbb{R}$, whereas no value is assigned to $Y_{i,j}$ on $\{1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0\}$. This perspective allows selection mechanisms to be incorporated into probabilistic statements without imputing counterfactual outcomes.

Definition 2.1 assumes a common latent aptitude distribution F_X across contexts, while allowing the observed achievement distributions F_{Y_i} to vary by system. This reflects a modeling choice in which the process generating latent aptitude is taken to be context-neutral, whereas the expression and measurement of aptitude may differ across contexts.

Direct evaluation of the order-statistic distributions in (2) is often cumbersome. However, under continuity, they admit a convenient representation in terms of Beta distributions. In particular,

$$F_{X_{i,(r_{i,j})}}\big(X_{i,(r_{i,j})}\big) = F_{U_{i,(r_{i,j})}} \circ F_X\big(X_{i,(r_{i,j})}\big)\,, \quad U_{i,(r_{i,j})} \sim \text{Beta}(r_{i,j},\, N_i - r_{i,j} + 1)\,, \tag{3}$$

and similarly,

$$F_{Y_{i,(j)}}\big(Y_{i,(j)}\big) = F_{U_{i,(j)}} \circ F_{Y_i}\big(Y_{i,(j)}\big)\,, \quad U_{i,(j)} \sim \text{Beta}(j,\, n_i - j + 1)\,. \tag{4}$$

This representation allows (2) to be computed via F_X, F_{Y_i}, and known Beta distributions, bypassing explicit order-statistic calculations. It also implies that latent aptitude can be recovered up to a monotone transformation without explicit knowledge of F_X.

Combining (3) and (4) yields

$$X_{i,(r_{i,j})} = F_X^{-1}\Big(F_{U_{i,(r_{i,j})}} \circ F_{U_{i,(j)}} \circ F_{Y_i}\big(Y_{i,(j)}\big)\Big)\,.$$

Define $W_{i,j} = F_{U_{i,(r_{i,j})}} \circ F_{U_{i,(j)}} \circ F_{Y_i}(Y_{i,(j)})$. Since F_X is monotone, so is its inverse, which implies

$$W_{i_1,j_1} \geq W_{i_2,j_2} \implies F_X^{-1}(W_{i_1,j_1}) \geq F_X^{-1}(W_{i_2,j_2}).$$

Thus the ordering of latent aptitudes is invariant to F_X, and F_X acts as a structural nuisance

parameter for comparative analysis. In practice, Full House Modeling can therefore be implemented via observed-to-rank mappings without requiring explicit specification of the latent distribution.

3 Paradoxes Under Identical Achievement Distributions

In this section, we derive explicit probabilities of the aptitude–achievement paradox, as defined in (1), within a Full House Modeling framework involving two populations. We focus on the setting in which the observed achievement distributions are identical across contexts, that is, $F_{Y_i} = F_Y$ for $i = 1, 2$ in Definition 2.1. The following definition formalizes the notion of paradox used throughout the remainder of the article.

Definition 3.1. *Consider the Full House Model in Definition 2.1 with two populations denoted by* $i = 1, 2$. *Let* $j \in \{1, \ldots, n_1\}$ *and* $k \in \{1, \ldots, n_2\}$ *denote fixed observed ranks in the two populations, where* n_i *is the number of observed individuals in population* i. *Thus,* $Y_{1,(j)}$ *and* $Y_{2,(k)}$ *are the* j*th and* k*th order statistics of the observed achievements in populations 1 and 2, respectively, and* $X_{1,(r_{1,j})}$ *and* $X_{2,(r_{2,k})}$ *are the corresponding latent aptitudes of those same individuals. We say that an aptitude-achievement paradox has occurred when both* $X_{1,(r_{1,j})} > X_{2,(r_{2,k})}$ *and* $Y_{1,(j)} < Y_{2,(k)}$ *have occurred. The aptitude-achievement paradox probability is defined as*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right). \tag{5}$$

As a simple illustration, consider the case in which the observed individuals in population i are exactly those with the top n_i latent aptitude values, so that $r_{i,t} = N_i - n_i + t$, for $t = 1, \ldots, n_i$. In particular, if we take $j = n_1$ and $k = n_2$, then $Y_{1,(n_1)}$ and $Y_{2,(n_2)}$ are the largest observed achievements in the two populations, and their corresponding latent aptitudes are

$$X_{1,(r_{1,n_1})} = X_{1,(N_1)}, \qquad X_{2,(r_{2,n_2})} = X_{2,(N_2)}.$$

If, moreover, $Y_{1,(n_1)} < Y_{2,(n_2)}$ a.s., then the achievement inequality in (5) holds automatically, so the aptitude–achievement paradox event reduces to $\{X_{1,(N_1)} > X_{2,(N_2)}\}$. Hence in this case,

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})},\, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right).$$

Under the conditions described below, this probability tends to 1 as $N_1 \to \infty$. This example shows that the paradox can arise even in a particularly simple setting where the observed individuals are the top latent-aptitude performers in each population.

We now introduce a convenient representation which allows the paradox probability to be expressed in terms of the latent aptitudes alone.

Lemma 3.2. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Then,*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right), \tag{6}$$

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(g_{j,k}\left(Y_{2,(k)}\right) < Y_{1,(j)} < Y_{2,(k)}\right) \tag{7}$$

where

$$h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot), \qquad g_{j,k}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}} \circ F^{-1}_{X_{2,(r_{2,k})}} \circ F_{Y_{2,(k)}}(\cdot).$$

Our first formal result shows that, under a monotonicity condition on the Beta-induced distortions, the paradox probability admits a closed-form expression.

Definition 3.3. *Let $\alpha, \beta > 0$. Then the regularized incomplete beta function is defined as*

$$I_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} \int_0^x u^{\alpha-1}(1-u)^{\beta-1} du,$$

where $B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta coefficient.

Theorem 3.4. *Consider the Full House Model in Definition 2.1 with two populations, where* $F_{Y_i} = F_Y$ *for* $i = 1, 2$. *Let* $j \in \{1, \ldots, n_1\}$ *and* $k \in \{1, \ldots, n_2\}$ *denote observed ranks, with corresponding latent ranks* $r_{1,j}$ *and* $r_{2,k}$. *Let* $I_{\alpha,\beta}$ *denote the regularized incomplete beta function as in Definition 3.3. Suppose that*

$$I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \;\geq\; 0, \quad \text{for all } t \in (0,1). \tag{8}$$

Then the paradox probability admits the explicit form

$$\begin{aligned}
&\mathrm{P}\big(X_{1,(r_{1,j})} > X_{2,(r_{2,k})},\, Y_{1,(j)} < Y_{2,(k)}\big) \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p,\, n_1+n_2-p-k+1)}{B(k, n_2-k+1)} - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p,\, N_1+N_2-p-r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

Remarks:

1. The paradox probability in Theorem 3.4 is distribution free. Once the Full House Model is specified, it depends only on the population sizes N_i, system sizes n_i, the system inclusion function, and the order-statistic indices. This follows from the Beta representations (3)–(4), which eliminate dependence on F_Y. In particular,

$$\begin{aligned}
h_{j,k}(x) &= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ \big(F_Y \circ F_Y^{-1}\big) \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x) \\
&= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x),
\end{aligned}$$

   so the dependence on F_Y cancels.

2. The latent aptitude distribution F_X likewise plays no role in the final expression. Although Lemma 3.2 represents the event in terms of latent order statistics, the monotone coupling induced by Full House Modeling ensures that all dependence on F_X cancels after transformation. Thus F_X acts as a structural nuisance parameter for paradox probabilities.

3. Combining these observations, the paradox probability admits the representation

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.$$

The positive-part operator reflects that the integrand may change sign. The technical condition (8) ensures that the integrand is nonnegative for all $t \in (0,1)$, allowing it to be removed and yielding the closed-form expression in Theorem 3.4. We characterize settings where this condition holds in subsequent results.

A natural case of interest is the comparison of individuals with extreme latent aptitudes across populations. The following corollary provides a closed-form expression for the paradox probability in this setting under a Full House Model with $F_{Y_i} = F_Y$ for $i = 1, 2$.

Corollary 3.4.1. *Suppose that the conditions of Theorem 3.4 hold and that the system inclusion function satisfies* $1_i(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1, i = 1, 2$, *that is, the highest-aptitude individual in each population is included in the observed system. Then,*

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \leq \frac{n_2}{N_2}, \\ 0, & \text{otherwise.} \end{cases}$$

Intuitively, this is the difference between two 'lottery' probabilities: the probability that individual 2 wins the observed achievement comparison (first term) minus the probability that individual 2 wins the latent aptitude comparison (second term). The paradox probability is positive precisely when the sampling ratio favors individual 2 more than the population ratio does. An analogous result holds for comparisons of individuals with the lowest latent aptitudes:

Corollary 3.4.2. *Suppose that the conditions of Theorem 3.4 hold and that the system inclusion function satisfies* $1_i(X_{i,(1)}, \mathbf{X}_{i,-(1)}) = 1, i = 1, 2$, *that is, the lowest-aptitude individual*

in each population is included in the observed system. Then,

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2}, \\ 0, & \text{otherwise.} \end{cases}$$

We now demonstrate a setting in which technical condition (8) holds. This setting is based on a configuration of (n_i, N_i) for $i = 1, 2$ under the assumption that the systems are populated with individuals from the source populations who have the highest aptitudes.

Proposition 3.5. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$, and let the system inclusion function be $1_i(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$. Denote by I the regularized incomplete beta function. Suppose that $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$, and set $k = j$. Under these assumptions, we conclude that technical condition (8) holds and the conclusion of Theorem 3.4 applies.*

The conditions in Proposition 3.5 suggest that the technical condition is driven by the relative comparison of sampling and population ratios. In particular, the requirement $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$ indicates that sampling distortion dominates population distortion in a coordinatewise sense. We conjecture that this condition can be relaxed to the single inequality $\frac{n_1}{n_2} \cdot \frac{N_2}{N_1} \leq 1$. Establishing this generalization would provide a sharper characterization of when the technical condition (8) holds. A proof of this result remains an open problem.

4 Paradox Under Different Distributions

In this section we will provide aptitude-achievement paradox probability results in a Full House Modeling setting involving two populations when we relax the specification that $F_{Y_1} = F_{Y_2}$ in Definition 2.1. We first provide bounds for aptitude-achievement paradox probability in general settings. We then provide explicit aptitude-achievement paradox probability results when Y_1 and Y_2 are normally distributed with different means and variances.

4.1 General bounds

The general bounds we derive require a relaxed form of first-order stochastic dominance under which the distributions of Y_1 and Y_2 are 'close' to each other. More formally, we introduce first-order ε-weak stochastic dominance, which is necessary for our aptitude-achievement paradox probability results.

Definition 4.1. *We will say that Y_2 is first order ε-weakly stochastically dominant over Y_1 if, for some $\varepsilon > 0$, $F_{Y_2}(y) \leq F_{Y_1}(y) \leq F_{Y_2}(y) + \varepsilon$ for all y, and there exists some y for which $F_{Y_2}(y) < F_{Y_1}(y)$.*

Remarks: The condition

$$F_{Y_2}(y) \leq F_{Y_1}(y) \qquad \text{for all } y$$

is exactly the usual first-order stochastic dominance condition [Quirk and Saposnik, 1962], so Y_1 is stochastically larger than Y_2. The additional upper bound

$$F_{Y_1}(y) \leq F_{Y_2}(y) + \varepsilon$$

does not reverse the dominance direction; rather, it requires that the pointwise gap between the two distribution functions be uniformly bounded by ε. Thus Y_2 dominates Y_1, but only by at most ε in the Kolmogorov distance on the one-sided scale. The strict inequality at some y excludes the degenerate case $F_{Y_1} \equiv F_{Y_2}$.

Under first-order ε-weak stochastic dominance, we derive a general lower and upper bound on the aptitude-achievement paradox probability when $F_{Y_1} \neq F_{Y_2}$.

Proposition 4.2. *Consider the Full House Model in Definition 2.1 with two populations $i = 1, 2$. Suppose that the system inclusion mechanism has the property that $1_i\left(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}\right) = 1$. Further suppose that Y_2 is first order ε-weakly stochastically dominant over Y_1 for some*

$\varepsilon > 0$ *and* $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$. *Then,*

$$\begin{aligned} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2} &\leq \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\leq n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2} \end{aligned} \tag{9}$$

Proposition 4.2 provides bounds on the paradox probability under general achievement distributions. For example, when $Y_i \sim N(\mu_i, \sigma^2)$, $i = 1, 2$, with $\mu_1 < \mu_2$ and common variance, ε can be bounded explicitly as a function of $(\mu_2 - \mu_1)/\sigma$. Beyond such bounds, we can obtain explicit aptitude-achievement paradox probabilities when $Y_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, 2$, allowing variances to differ across contexts.

4.2 Normal distributions

This section develops explicit aptitude-achievement paradox probabilities (5) when a Full House Model in Definition 2.1 specifies $Y_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, 2$. We focus on systems where $1_i(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$ for $i = 1, 2$. The outline is similar to Section 3: a technical condition is needed to prove the main result, and further results provide configurations satisfying this condition. However, with $Y_i \sim N(\mu_i, \sigma_i^2)$, the technical condition yields a remainder term that prevents an exact closed-form expression. We show that this remainder is negligible under mild conditions on population fractions and distributional parameters.

Similar to the proof of Theorem 3.4, Lemma 3.2 will provide a representation more conducive to formal integration needed to prove our main result. Only this time we will use (7) in Lemma 3.2 to represent the aptitude-achievement paradox probability as

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right),$$

where, from the definition of g in Lemma 3.2 applied to our current setting, we have

$$g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2N_1}{N_2n_1}}\right)\sigma_1 + \mu_1, \tag{10}$$

and Φ defines the cumulative distribution function of the standard normal distribution. From here, we have that

$$\begin{aligned}
&\mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2N_1}{N_2n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\quad = \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y)dF_{Y_2,(n_2)}(x),
\end{aligned}$$

where $A := A(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$ is the range of integration so that the inner integral above is properly specified. More formally,

$$\begin{aligned}
A &= \left\{\omega : F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)}(\omega))^{\frac{n_2N_1}{N_2n_1}}\right) \le Y_{2,(n_2)}(\omega)\right\} \\
&= \left\{\omega : \Phi\left(\frac{Y_{2,(n_2)}(\omega)-\mu_2}{\sigma_2}\right)^{\frac{n_2}{N_2}} \le \Phi\left(\frac{Y_{2,(n_2)}(\omega)-\mu_1}{\sigma_1}\right)^{\frac{n_1}{N_1}}\right\},
\end{aligned} \tag{11}$$

where the second equality in (11) follows from (10). Continuing with the probability calculation above

$$\begin{aligned}
&\int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y)dF_{Y_2,(n_2)}(x) \\
&= \int_A F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x) - \int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x).
\end{aligned} \tag{12}$$

The left-hand side of (12) can (and will) be further evaluated and leads to the desired explicit aptitude-achievement paradox probability. The right-hand side of (12) represents a

remainder term $R := R(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$,

$$R = -\int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_{2,(n_2)}}(x), \tag{13}$$

where the set A in (13) is defined in (11). The remainder R is negligible when n_2 is moderately large, as we now show. From (11) and (13), we have

$$\begin{aligned}
|R| &= \left| \int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_{2,(n_2)}}(x) \right| \\
&\leq \int_{A^c} \left| F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} \right| dF_{Y_{2,(n_2)}}(x) \\
&\leq \mathrm{P}\left(\Phi\left(\frac{Y_{2,(n_2)} - \mu_2}{\sigma_2} \right)^{\frac{n_2}{N_2}} \geq \Phi\left(\frac{Y_{2,(n_2)} - \mu_1}{\sigma_1} \right)^{\frac{n_1}{N_1}} \right).
\end{aligned}$$

Now, when we specify that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$ as in Proposition 4.2, we have

$$\begin{aligned}
&\mathrm{P}\left(\Phi\left(\frac{Y_{2,(n_2)} - \mu_2}{\sigma_2} \right)^{\frac{n_2}{N_2}} \geq \Phi\left(\frac{Y_{2,(n_2)} - \mu_1}{\sigma_1} \right)^{\frac{n_1}{N_1}} \right) \quad \leq \mathrm{P}\left(\Phi\left(\frac{Y_{2,(n_2)} - \mu_2}{\sigma_2} \right) \geq \Phi\left(\frac{Y_{2,(n_2)} - \mu_1}{\sigma_1} \right) \right) \\
&\quad = \mathrm{P}\left(\frac{Y_{2,(n_2)} - \mu_2}{\sigma_2} \geq \frac{Y_{2,(n_2)} - \mu_1}{\sigma_1} \right).
\end{aligned}$$

Under the assumption that $\sigma_1 \leq \sigma_2$, straightforward algebra yields

$$\mathrm{P}\left(\frac{Y_{2,(n_2)} - \mu_2}{\sigma_2} \geq \frac{Y_{2,(n_2)} - \mu_1}{\sigma_1} \right) = \mathrm{P}\left(Y_{2,(n_2)} \leq \left(\frac{\mu_2}{\sigma_2} - \frac{\mu_1}{\sigma_1} \right) \left(\frac{1}{\sigma_2} - \frac{1}{\sigma_1} \right)^{-1} \right).$$

Under the additional assumption that $\mu_1 \leq \mu_2$, the right-hand side is bounded above by $\mathrm{P}\left(Y_{2,(n_2)} \leq \mu_2\right)$. Putting it all together, we have $|R| \leq \mathrm{P}\left(Y_{2,(n_2)} \leq \mu_2\right) \leq \left(\frac{1}{2}\right)^{n_2}$, when $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $\sigma_1 \leq \sigma_2$, and $\mu_1 \leq \mu_2$. Therefore, although the remainder term in (13) arises from the restriction of the integration domain to A, it is exponentially small in n_2 under a familiar ordering of population fractions (see Proposition 4.2) and a simple parametric configuration. Consequently, the aptitude–achievement paradox probability is well approximated by the unrestricted integral given by the left-hand side of (12)with R contributing only a negligible

correction for moderately large n_2.

We now introduce a multivariate normal representation that allows this principal term to be expressed in closed form. Let $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ and covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}, \tag{14}$$

defined with the following structure

$$\begin{aligned} \Sigma_{11} &= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1 \times n_1} \quad \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1)\times n_1} \\ \Sigma_{22} &= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} . \end{aligned} \tag{15}$$

With this structure we now provide an explicit aptitude-achievement paradox probability, the main result of this section.

Theorem 4.3. *Assume a Full House Model in Definition 2.1 and additionally specify that* $Y_i \sim N(\mu_i, \sigma_i^2)$*, and suppose that the system inclusion mechanism has the property that* $1_i(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}) = 1$ *for* $i = 1, 2$*. Let* $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ *where* $\mathbf{\Sigma}$ *is as defined in* (14) *and* (15)*. Then,*

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad = n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \dots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \dots, 0)) - \frac{N_2}{N_1+N_2} + R, \end{aligned} \tag{16}$$

where R is defined in (13), $F_{\mathbf{W}}$ is the distribution function of $\mathbf{W}$, and the first n_1 elements of $\mathbf{w}$ are $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last n_2-1 elements are 0.

Moreover, when $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $\sigma_1 \leq \sigma_2$, and $\mu_1 \leq \mu_2$, we have that

$$|R| \leq \left(\frac{1}{2}\right)^{n_2}. \tag{17}$$

We now show that plug-in estimators of the normal distribution parameters μ_i and σ_i, $i = 1, 2$, yield reliable estimates of aptitude-achievement paradox probabilities (16).

Theorem 4.4. *Assume a Full House Model in Definition 2.1 and additionally specify that $Y_i \sim N(\mu_i, \sigma_i^2)$, and suppose that the system inclusion mechanism has the property that $1_i(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}) = 1$ for $i = 1, 2$. Let $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \boldsymbol{\Sigma})$ where $\boldsymbol{\Sigma}$ is as defined in (14) and (15). Assume the parameters n_1, n_2, μ_1, μ_2, σ_1, and σ_2 satisfy the assumptions in Theorem 4.3, and as $n_1, n_2 \to \infty$, the relative growth rates of the dimensions satisfy*

$$\lim_{n_1,n_2\to\infty} \frac{\ln n_2}{\ln n_1} > \left(\frac{\sigma_1}{\sigma_2}\right)^2.$$

In addition, the μ_3, μ_4, σ_3, and σ_4 satisfy the same assumptions as μ_1, μ_2, σ_1, and σ_2 above,

then we have

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_3, \mu_4, \sigma_3, \sigma_4\right) \\ -\,&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_1, \mu_2, \sigma_1, \sigma_2\right) \to 0 \end{aligned}$$

as $n_1, n_2 \to \infty$.

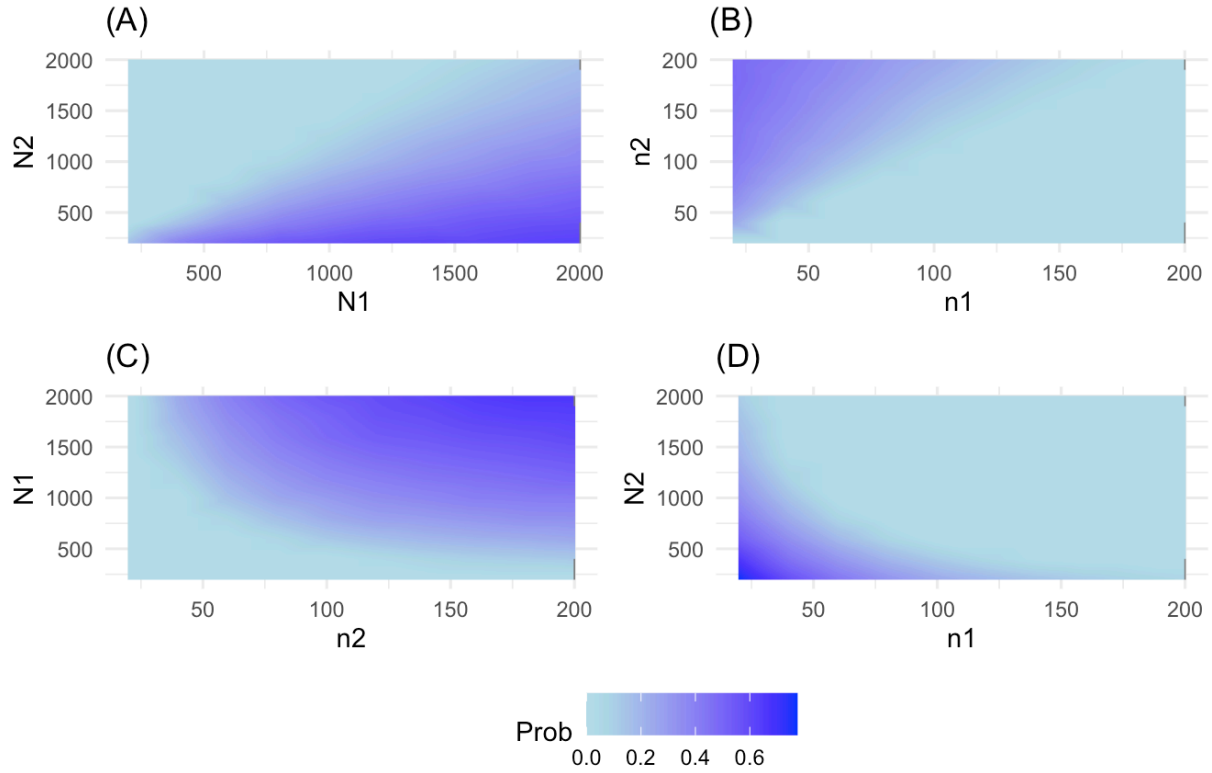


Figure 1: Sensitivity Analysis for Paradox Probability

5 Examples

5.1 Sensitivity Analysis for Theorem 3.4

Under the setting of Theorem 3.4, we will specify $X_1 \overset{d}{=} X_2 \overset{d}{=} X$, $Y_1 \overset{d}{=} Y_2 \overset{d}{=} Y$, where $X \sim \text{Pareto}(\alpha)$ and $Y \sim \mathcal{N}(0,1)$. Without loss of generality, we set $j = n_1 - 1$ and $k = n_2 - 1$, corresponding to the selection of the second-best performer in the aptitude-achievement paradox probability, as defined in Definition 3.1.

In each simulation, we randomly sample n_1 and n_2 observations from the distribution of Y, and compute their latent aptitudes using Equation (2). Based on these values, we directly calculate paradox probabilities according to Definition 3.1. We repeat this simulation 10000 times and report the average.

Figure 1 presents the results of the sensitivity analysis for the aptitude-achievement

paradox probability described in Theorem 3.4. In Figure 1 (A), we fix $n_1 = 40$ and $n_2 = 60$, and vary N_1 and N_2 from 200 to 2000. In Figure 1 (B), we fix $N_1 = 10^3$ and $N_2 = 10^3$, and vary n_1 and n_2 from 20 to 200. In Figure 1 (C), we fix $n_1 = 40$ and $N_2 = 10^3$, and vary n_2 from 20 to 200 and N_1 from 200 to 2000. In Figure 1 (D), we fix $N_1 = 10^3$ and $n_2 = 60$, and vary N_2 from 200 to 2000 and n_1 from 20 to 200.

As shown in Figure 1 (A), a small N_2 and large N_1 increase the paradox probability by making $\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}\right) \to 1$. Similarly, in Figure 1 (B), a small n_1 combined with a large n_2 leads to a higher aptitude-achievement paradox probability since it drives $\mathrm{P}\left(Y_{1,(j)} < Y_{2,(k)}\right) \to 1$.

5.2 Ratio of N_2 and N_1

Here we investigate the effect of the ratio between N_2 and N_1 on aptitude-achievement paradox probabilities under the specifications of Corollary 3.4.1. Let $c = \frac{N_2}{N_1}$, and vary c from 0.1 to 1 in increments of 0.1. In each simulation, we fix $n_1 = 60$ and $n_2 = 40$, and vary N_1 from 100 to 2000. We repeat this simulation 10000 times and report the average paradox probability.

Figure 2 presents the results. As shown in the figure, the paradox probability decreases as c increases, which is reasonable since $\mathbb{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}\right)$ decreases as N_2 increases. Importantly, Figure 2 also demonstrates that the paradox probability remains constant when the ratio between N_2 and N_1 is fixed. This implies that it is unnecessary to focus on the exact values of N_1 and N_2 when computing the paradox probability; rather, the ratio N_2/N_1 alone determines its value.

5.3 Recursive Full House Models

In this section we will investigate a simple extension to Full House Modeling in which Full House Models are repeatedly applied to individuals as they progress through iterations of some generic process. The general setup is that, in the first round, latent aptitude $X_j \overset{iid}{\sim}$

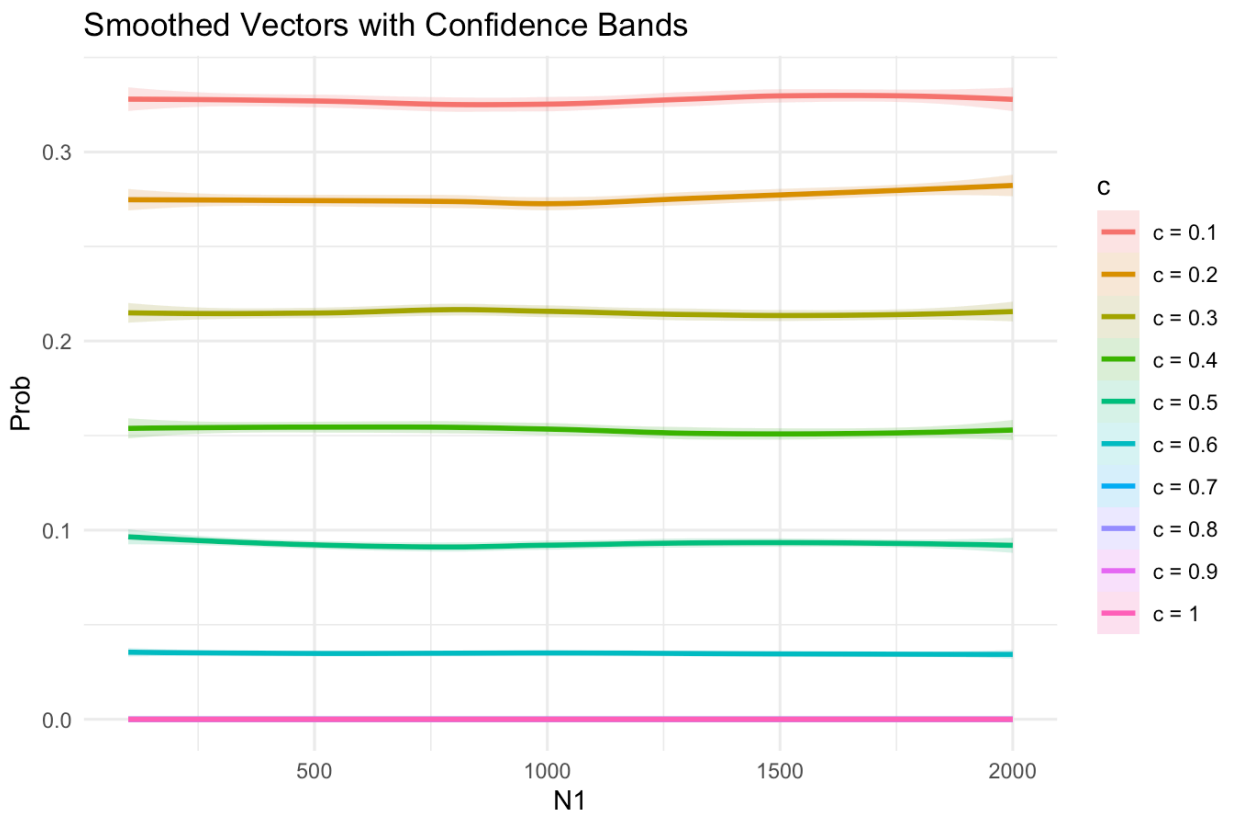


Figure 2: Effect of the ratio between N_2 and N_1 on aptitude-achievement paradox probabilities under the specifications of Corollary 3.4.1.

F_X, $j = 1, \ldots, N$ will be generated and then a label will be randomly assigned to each individual. There will be m labels and each label will designate the outcome distribution. $F_{Y_{i,j}}$, $i = 1, \ldots, n$, $j = 1, \ldots, n_i$. Then outcomes will be realized as

$$Y_{i,(r_{i,(j)})} = F^{-1}_{Y_{i,j}} \circ I^{-1}_{r_{i,(j)}, n_i - r_{i,(j)}+1} \circ I_{j,N-j+1} \circ F_X(X_{(j)}), \tag{18}$$

where $r_{i,(j)}$ is the rank of $X_{(j)}$ among the n_i latent aptitudes also assigned label i.

In the next iteration, all observations will be combined and then sorted in descending order by $Y_{i,(j)}$. The labels will be assigned in descending order, the largest n_m values of $Y_{i,(j)}$ are assigned label m, the next largest n_{m-1} values of $Y_{i,(j)}$ are assigned label $m-1$, and so on. We will setup these simulations so that Y_i stochastically dominates Y_j whenever $i > j$ so that higher outcomes are associated with higher labels. Then new Y values are generated

according to (18). The process will repeat until a pre-assigned stopping rule is satisfied. Note that this simulation setup assumes an immutable latent aptitude.

5.3.1 Stop when highest latent aptitude is among the highest-performing group

Here we consider a recursive Full House Modeling setup described in Section 5.3 in which the recursion stops at the first iteration when $X_{(N)}$ is assigned label m. Here we specify that $N = 100,000$, $m = 200$, $n_i = 500$, $i = 1, \ldots, m$. We will assume that $X_j \overset{iid}{\sim}$ Pareto(1.16), $j = 1, \ldots, N$. And we will assume that $Y_{i,j} \overset{iid}{\sim} N(\mu_i, \sigma_i)$, $j = 1, \ldots, n_i$ and $i = 1, \ldots, m$. We will specify that the μ_is as a set of 200 equally spaced values between 60 and 80 with μ_i increasing in i, ie $\mu_i = 60 - 20/199 + 20i/199$, $i = 1, \ldots, m$. We will fix $\sigma_i = \sigma$ and consider $\sigma \in \{0.5, 1, 2\}$. These simulations will be repeated 500 times.

To illustrate the procedure, consider a simplified scenario in which all $X_{(j)}$ values are realized at their median value. With this simplification, $Y_{i,(r_{i,(j)})}$ in (18) is approximately $F^{-1}_{Y_{i,j}}((r_{i,(j)} - 1/3)/(n_i + 1/3))$ [Kerman, 2011]. And we have that $Y_{i,(n_i)} - Y_{i+1,(n_{i+1}-1)} > \mu_{i+1} - \mu_i$ for all considered values of σ. The individual with latent ability $X_{(N)}$ will consistently attain the value $Y_{i,(n_i)}$, regardless of their assigned group. Consequently, this individual's position among the m groups will improve as the simulation advances.

Results can be seen in Figure 3. These Monte Carlo simulations reveal that it typically takes several rounds before the individual with the top aptitude belongs to the group that allows for the highest performances. On average, with σ set to 0.5, 1, and 2, it takes 2.61, 4.19, and 7.46 rounds, respectively, for the individual with the top aptitude to join the highest-performing group (respective standard errors: 0.05, 0.09, 0.18). Deriving the distribution of this hitting time analytically under explicit FHM assumptions, and characterizing how it depends on σ and grouping structure, remains an open problem.

While the setting is stylized and assumes fixed latent aptitude, it illustrates how structural features of evaluative systems, not just randomness, can delay recognition of true potential. Whereas the paradox probability quantifies cross-sectional reversal risk, recursive Full House

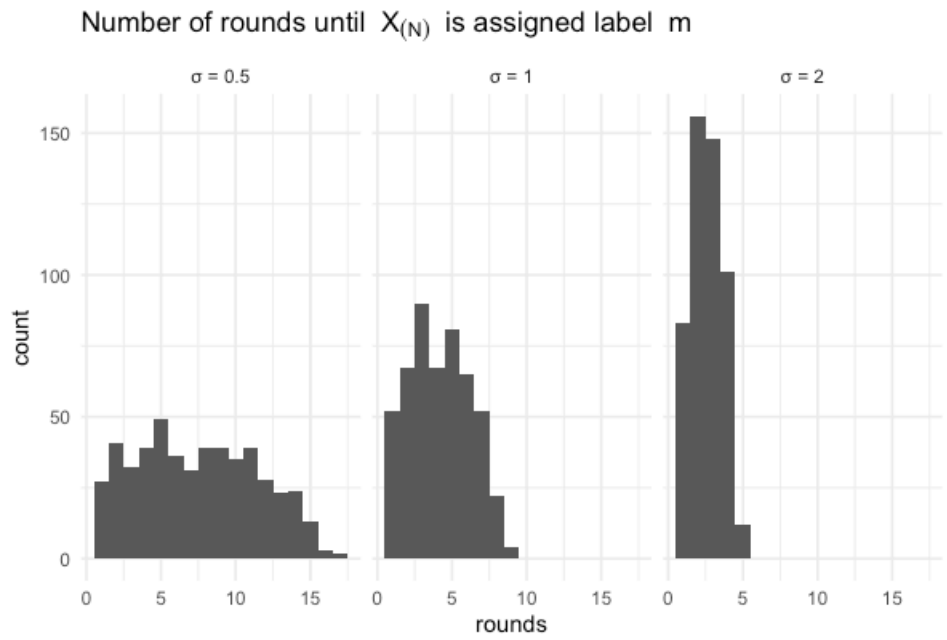


Figure 3: The number of rounds that it takes for $X_{(N)}$ to be assigned label m in a recursive Full House Modeling framework in Section 5.3 with specifications in Section 5.3.1.

Modeling captures temporal persistence: the hitting time measures how many rounds elapse before latent aptitude and observed achievement align. These findings resonate with a growing empirical literature on bias and distortion in algorithmic and educational evaluation systems, which we discuss further in the Discussion.

5.4 Batting Averages in Major League Baseball

We now apply Theorem 4.3 and the reliability guarantees in Theorem 4.4 to evaluate the contextual extremeness of observed outcomes in a real-world system where the underlying distribution of aptitude is not directly observable and evolves over time. Major League Baseball (MLB) offers a canonical example of such a system. Here, player skill (latent batting aptitude) is inferred only indirectly through season-level observed outcomes (batting averages), which are shaped by changing population structures and environmental variance. In particular, MLB introduced a series of rule changes in 2023 intended to increase in-game action, including a restriction on defensive positioning designed to raise batting averages

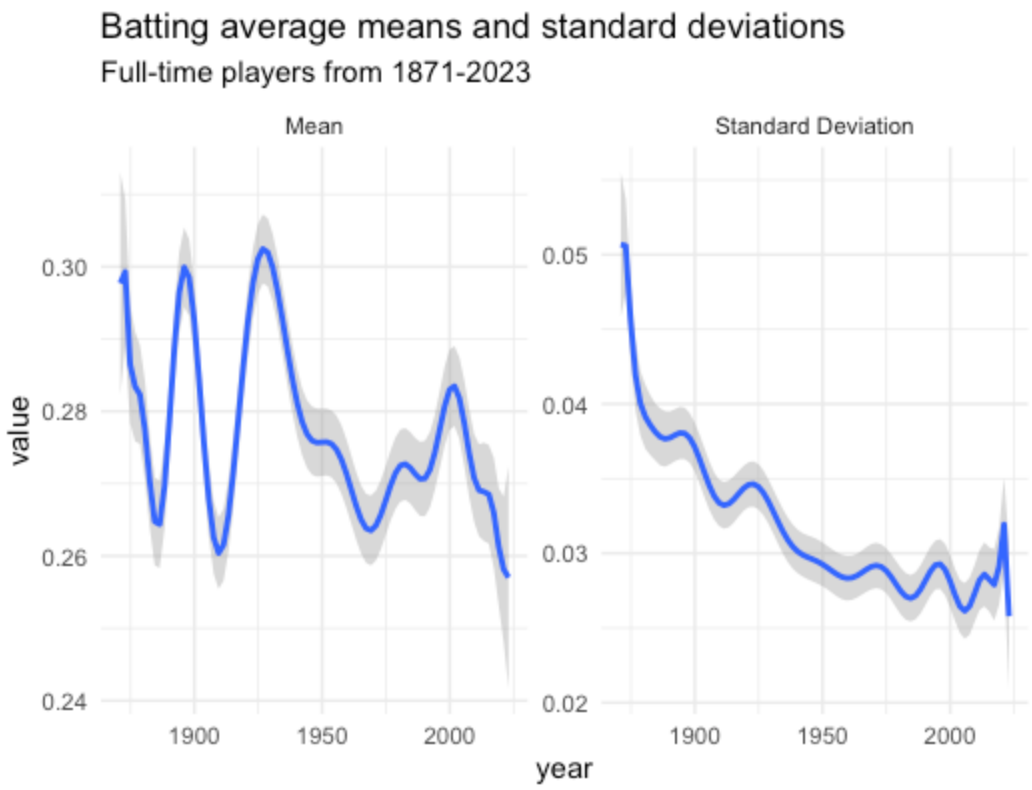


Figure 4: Seasonal means and standard deviations of batting averages for full-time players from 1871-2023. Smoothing is applied to both trends.

on balls in play (BABIP) [Castrovince, 2023]. Figure 4 presents smoothed estimates of the mean and standard deviation of batting averages for full-time players from 1871 to 2023.

The 2023 season saw an increase in mean batting average for full-time players of 0.0057 (corresponding to an additional hit about every four baseball games) with a reduction in variability. A two-sample test for differences in batting averages among full-time players from the 2023 and 2022 seasons has a p-value of 0.039. Thus, rule changes seemed to be effective in yielding increased batting averages[1] Moreover, the 2023 MLB batting leader Luis Arraez had a 0.354 batting average which was the first batting average to exceed .350 (a mark of excellence) since Josh Hamilton in 2010.

We now apply Theorem 4.3 to quantify paradox probabilities for extreme batting aver-

[1]Technically, Castrovince [2023] referred to MLB's desire to increase batting averages on balls in play (BABIP) while this analysis focused on batting averages. BABIP increased by 0.0061 from 2022 to 2023.

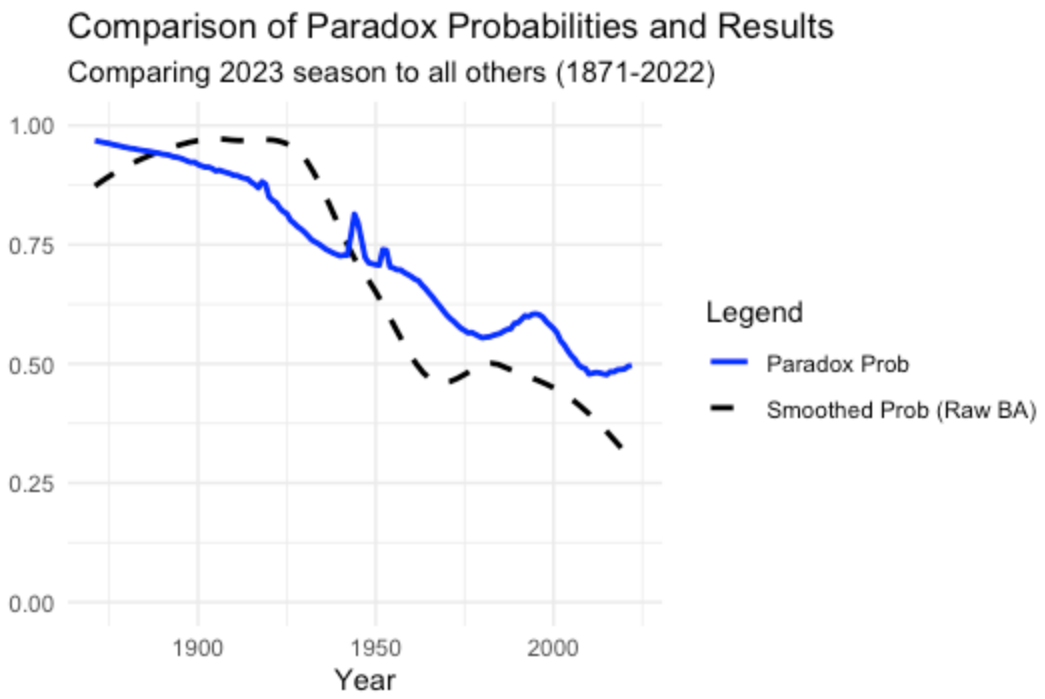


Figure 5: Comparison of paradox probabilities and empirical exceedance rates relative to the 2023 season, based on data from 1871–2022. The blue line shows estimated paradox probabilities via plug-in using Theorem 4.3. The black dashed line displays a smoothed empirical exceedance rate for Arraez's 2023 batting average.

ages, treating the 2023 season as system 1 and each season from 1871 to 2022 as system 2. Seasonal batting averages of full-time players are assumed normally distributed, with means and standard deviations estimated via plug-in; the reliability guarantees of Theorem 4.4 apply. Figure 4 shows that both the average and variability of batting averages in 2023 are among the lowest observed, indicating a historically suppressed environment. Moreover, with our specifications we have that $\hat{\sigma}_1 \leq \hat{\sigma}_2$, $\hat{\mu}_1 \leq \hat{\mu}_2$, and $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, so the remainder term (13) in Theorem 4.3 is negligible. The talent pool estimates N_i are taken from Yan et al. [2025], which includes sensitivity analyses showing that qualitative conclusions are robust to reasonable perturbations of these estimates.

Figure 5 displays the resulting paradox probabilities alongside a smoothed empirical exceedance rate, the proportion of historical seasons in which the maximum batting average exceeded Arraez's 2023 mark. The estimated paradox probabilities closely track this bench-

mark. Both quantities increase as one moves backward in time, reflecting that earlier eras were structurally more conducive to extreme observed batting averages due to higher means, larger variances, and smaller effective populations.

Under Full House Modeling, Arraez's 2023 performance ranks 68th all-time among full-time batters after accounting for population depth and contextual effects [Yan et al., 2025, Eck et al., 2025], in contrast to his raw observed rank of 341. This discrepancy illustrates how observed dominance can be distorted by changes in population structure and measurement environments. Paradox probabilities provide a formal way to quantify this distortion, clarifying why exceptional modern performances may appear historically unremarkable when evaluated without contextual adjustment.

6 Discussion

The aptitude-achievement paradox arises when an individual from one context demonstrates higher latent aptitude but simultaneously lower observed achievement than an individual from another context. Although widely encountered in education, labor economics, psychometrics, medicine, and sports analytics, measuring the frequency of this paradox under realistic data-generating regimes is nontrivial. In this article, we formalize the Full House Methodology and investigate the probabilities of observing paradoxes between latent aptitude and observed achievement in comparative analysis of individuals across contexts.

This work is a theoretical clarification of Full House Modeling and provides the definition and solutions for an inferential target, the aptitude-achievement paradox, which extend beyond the baseball motivations in Yan et al. [2025]. In doing so, we note the baseball origins of this work, as articulated by Stephen Jay Gould in his 1996 book *Full House: The Spread of Excellence from Plato to Darwin* (p. 122), from which this framework takes its name: "Baseball is a truly remarkable system for statisticians, manifesting two properties devoutly to be wished, but not often encountered, in actual data: an institution that has

worked by the same rules for a century, and has compiled complete data (nothing major missing) on all measurable aspects of its history." In this way, baseball, with its century of consistent rules and comprehensive measurement, served as an ideal proving ground for the Full House framework [Yan et al., 2025]. Gould concluded his reflection on baseball's statistical regularity with the admission that his sense of calculating "something quite general about the nature of systems" was "just a feeling, not a proof." Here, our work offers more than a feeling, providing rigorous proofs that yield explicit and easy-to-calculate aptitude-achievement paradox probabilities.

We derive closed-form expressions for paradox probabilities under Full House data generating models in two regimes: (i) when observed achievements are i.i.d., and (ii) when context-specific summaries are independent normals with distinct means and variances. The probability rises with population-size asymmetry (large N_1, small N_2) and sampling asymmetry (small n_1, large n_2). Moreover, for a fixed population-size ratio N_2/N_1, it is unchanged by proportional rescaling of the sample sizes (n_1, n_2). We also develop bounds for more general settings and show that, when observed achievement is normal, plug-in estimators yield reliable paradox-probability estimates.

Beyond baseball, the recursive Full House Modeling example in Section 5.3.1 speaks to broader concerns in educational evaluation and admissions. When students transition across institutions of varying selectivity, from less selective undergraduate programs to more selective graduate programs, or vice versa, observed performance metrics may systematically misrepresent underlying aptitude due to differences in both competition depth and measurement environments. The framework developed here provides a principled way to quantify such distortions, though substantial work remains to characterize the specific selection mechanisms and measurement properties that govern real educational pipelines. Our hope is that this methodology can eventually inform more equitable evaluation practices across educational and professional contexts where cross-institutional comparisons are routine but poorly understood.

By elevating aptitude-achievement paradoxes from an anecdotal concern to a model-based, estimable quantity, our framework clarifies when cross-context comparisons are trustworthy, when they are fragile, and how to report that fragility responsibly. This closes the loop between principled latent measurement, realistic selection processes, and interpretable comparative claims.

Acknowledgements

We are grateful to ... and many others for helpful comments, great discussions, and inspiration. We acknowledge the use of ChatGPT-5.2 and Claude 4.5 to aid with proofreading and writing clarity.

Supplementary Materials

The Supplementary Materials contain proofs of mathematical results, together with auxiliary Monte Carlo simulations provided solely for numerical intuition for Theorem 4.3.

References

David J Bartholomew, Martin Knott, and Irini Moustaki. *Latent variable models and factor analysis: A unified approach*. John Wiley & Sons, 2011.

Benjamin S Baumer and Gregory J Matthews. A statistician reads the sports page: there is no avoiding WAR. *Chance*, 27(3):41–44, 2014.

Benjamin S Baumer, Shane T Jensen, and Gregory J Matthews. openwar: An open source system for evaluating overall player performance in major league baseball. *Journal of Quantitative Analysis in Sports*, 11(2):69–84, 2015.

Scott M Berry, C Shane Reese, and Patrick D Larkey. Bridging different eras in sports. *Journal of the American Statistical Association*, 94(447):661–676, 1999.

Trevor G Bond and Christine M Fox. *Applying the Rasch model: Fundamental measurement in the human sciences*. Psychology Press, 2013.

Ralph A. Bradley. Science, statistics, and paired comparisons. *Biometrics*, 32(2):213–239, 1976. doi: 10.2307/2529494.

Ralph A. Bradley. Paired comparisons: Some basic procedures and examples. In P. R. Krishnaiah and P. K. Sen, editors, *Handbook of Statistics*, volume 4, pages 299–326. Elsevier, Amsterdam, 1984.

Ralph A. Bradley and Milton E. Terry. The rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39:324–345, 1952. doi: 10.1093/biomet/39.3-4.324.

Anthony Castrovince. All of your questions about the new baseball rules, answered. *MLB.com*, February 2023. URL https://www.mlb.com/news/mlb-new-rules-for-2023-faq.

Wei Chu and Zoubin Ghahramani. Preference learning with gaussian processes. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 137–144. ACM, 2005. doi: 10.1145/1102351.1102369.

Herbert A. David. *The Method of Paired Comparisons*. Chapman and Hall, London, 2 edition, 1988.

Daniel Eck, Jun Hee Kim, Shen Yan, Muskan Bhatla, and Zheer Wang. *fullhouse: Package for Estimating Talent and Era-Adjusted Performance Metrics*, 2025. URL https://github.com/DEck13/fullhouse. R package version 1.1.

Bradley Efron and Carl Morris. Data analysis using stein's estimator and its generalizations. *Journal of the American Statistical Association*, 70(350):311–319, 1975.

Arpad E. Elo. *The Rating of Chessplayers, Past and Present*. Arco Publishing, 1978.

Susan E Embretson and Steven P Reise. *Item response theory for psychologists*. Psychology Press, 2000.

Thore Graepel, Tom Minka, and R TrueSkill Herbrich. A bayesian skill rating system. *Advances in Neural Information Processing Systems*, 19(569-576):7, 2007.

Jacques A Hagenaars and Allan L McCutcheon. *Applied latent class analysis*. Cambridge University Press, 2002.

Paul W Holland and Howard Wainer. *Differential item functioning*. Routledge, 2012.

Neil Houlsby, Ferenc Huszar, Zoubin Ghahramani, and Jose Hernández-lobato. Collaborative gaussian processes for preference learning. *Advances in neural information processing systems*, 25, 2012.

Guido Imbens and Konrad Menzel. A causal bootstrap. Technical report, National Bureau of Economic Research, 2018.

Maurice G Kendall. A new measure of rank correlation. *Biometrika*, 30(1-2):81–93, 1938.

Jouni Kerman. A closed-form approximation for the median of the beta distribution. *arXiv preprint arXiv:1111.0433*, 2011.

Frederic M Lord. A paradox in the interpretation of group comparisons. *Psychological bulletin*, 68(5):304, 1967.

Roderick P McDonald. *Test theory: A unified treatment*. psychology press, 1999.

Stephen Nickell and Richard Layard. Labor market institutions and economic performance. *Handbook of labor economics*, 3:3029–3084, 1999.

James P Quirk and Rubin Saposnik. Admissibility and measurable utility functions. *The Review of Economic Studies*, 29(2):140–146, 1962.

Stephen Raudenbush and Anthony S Bryk. A hierarchical model for studying school effects. *Sociology of education*, pages 1–17, 1986.

Michael J Schell. *Baseball's all-time best hitters*. Princeton University Press, 2013.

Michael J Schell. *Baseball's all-time best sluggers*. Princeton University Press, 2016.

Edward H Simpson. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 13(2):238–241, 1951.

Robert H Somers. A new asymmetric measure of association for ordinal variables. *American sociological review*, pages 799–811, 1962.

Wayne L Winston, Scott Nestler, and Konstantinos Pelechrinis. *Mathletics: How Gamblers, Managers, and Fans use Mathematics in Sports*. Princeton University Press, 2022.

Shen Yan, Adrian Burgos, Jr, Christopher Kinson, and Daniel J Eck. Comparing baseball players across eras via novel full house modeling. *The Annals of Applied Statistics*, 19(2): 1778–1799, 2025.

Amy Zhao Yu, Shahar Ronen, Kevin Hu, Tiffany Lu, and César A Hidalgo. Pantheon 1.0, a manually verified dataset of globally famous biographies. *Scientific data*, 3(1):1–16, 2016.

The representation in (2) may appear as a modeling choice. The following result shows that, under natural invariance and rank-based assumptions, this construction is in fact essentially unavoidable.

Theorem 6.1 (Characterization of Full House Modeling)**.** *Consider a class of models generating paired variables* $(X_{i,j}, Y_{i,j})$ *under the following conditions:*

1. ***Latent exchangeability.*** *For each population* i*, the latent variables* $X_{i,1}, \ldots, X_{i,N_i}$ *are i.i.d. draws from a continuous distribution* F_X*.*

2. ***Rank-based selection.*** *Inclusion in the observed system is determined by a measurable function of latent ranks:*

$$1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 1 \iff r_{i,j} \in \mathcal{R}_i,$$

for some subset $\mathcal{R}_i \subset \{1, \ldots, N_i\}$.

3. ***Rank preservation.*** *For individuals included in the system, observed outcomes are strictly increasing functions of latent aptitude:*

$$X_{i,(r)} > X_{i,(s)} \implies Y_{i,(r)} > Y_{i,(s)}.$$

4. ***Distributional invariance.*** *The joint model is invariant under strictly increasing transformations of latent aptitude, i.e., for any strictly increasing function* h,

$$\{X_{i,j}, Y_{i,j}\} \overset{d}{=} \{h(X_{i,j}), Y_{i,j}\}.$$

Then, for each population i *and each observed rank* j, *there exists a representation*

$$Y_{i,(j)} = F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}}\left(X_{i,(r_{i,j})}\right),$$

that is, the observed outcomes arise from a quantile-matching (rank-preserving) transformation of latent aptitude.

Moreover, any model satisfying (1)–(4) is equivalent, in distribution of (X, Y) *pairs among selected individuals, to a Full House Model as defined in Definition 2.1.*

Proof sketch. The argument proceeds in three steps.

Step 1: Reduction to ranks. By exchangeability and continuity of F_X, the latent sample induces a uniform distribution over ranks. Thus, conditional on selection, all relevant information about $X_{i,(r)}$ is captured by its rank r.

Step 2: Monotone representation. Rank preservation implies that, for selected individuals, $Y_{i,(j)}$ is a strictly increasing function of $X_{i,(r_{i,j})}$. Hence there exists a measurable, strictly increasing function ψ_i such that

$$Y_{i,(j)} = \psi_i\big(X_{i,(r_{i,j})}\big)\,.$$

Step 3: Invariance implies quantile coupling. Distributional invariance under strictly increasing transformations of X implies that ψ_i cannot depend on the scale of X, but only on its relative position within the population. Therefore ψ_i must act through the cumulative distribution function:

$$\psi_i(x) = \tilde{\psi}_i(F_{X_{i,(r_{i,j})}}(x)).$$

Matching marginal distributions of $Y_{i,(j)}$ then uniquely determines $\tilde{\psi}_i$ as the inverse CDF of $Y_{i,(j)}$, yielding

$$Y_{i,(j)} = F^{-1}_{Y_{i,(j)}}\Big(F_{X_{i,(r_{i,j})}}(X_{i,(r_{i,j})})\Big)\,.$$

Step 4: Equivalence. This representation coincides with the Full House Model construction in Definition 2.1. Since all models satisfying (1)–(4) admit this representation, they are equivalent in distribution over observed (X, Y) pairs. □